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GD2-specific CAR T cells encapsulated in an injectable hydrogel control retinoblastoma and preserve vision

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Abstract

Retinoblastoma (RB) is a pediatric retinal tumor that overexpresses the ganglioside GD2. Although it is treatable in patients with early diagnosis, patients may lose one or two eyes. We generated GD2-specific chimeric antigen receptor T lymphocytes (GD2.CAR-Ts) and locally delivered them to mice with an in-situ grafting RB. When used in combination with the local release of interleukin (IL)-15 and an injectable hydrogel, we showed that GD2.CAR-Ts successfully eliminate RB tumor cells without impairment of the mouse vision.

Introduction

RB is a pediatric cancer developed within the retina that occurs in both heritable form in association with germ-line *RB1* gene (encodes RB transcriptional corepressor 1) mutation or more frequently in nonheritable form¹. When RB is confined to the eye, cryotherapy, thermotherapy, chemotherapy and radiotherapy are used, but enucleation, with consequent

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Author contributions statement

K. W., G. D., B. S., and Z. H. designed the project. K. W., Y. C., S. A., M. Z., and E. L. conducted the experiments and wrote the manuscript. K. W., Y. C., S. A., M. Z., E. L., B. S., and Z. H. contributed the protocols and analyzed the data. G. D., B. S. and Z. H. edited the grammar and critically reviewed the manuscript. G. D., B. S. and Z. H. supervised the project. The manuscript was reviewed by all the authors.

Competing Interests statement

The authors declare no conflicts of interest.

loss of vision, is frequently the best option for advanced stages^{2,3}. Development of treatment strategies that can eradicate the tumor, but preserve the vision would significantly enhance the quality of life of RB patients. Immunotherapy in the form of adoptive transfer of CAR-Ts is perceived as the most promising therapy for pediatric patients with acute lymphoblastic leukemia⁴. Here we have developed a local immunotherapy approach based on GD2-specific CAR-Ts releasing IL-15 and encapsulated in *in-situ* formed injectable chitosan hydrogel that allows tumor eradication whilst preserving the vision.

RB like other neuroepithelial cancer expresses the ganglioside GD2⁵. GD2 has been targeted in pediatric patients with neuroblastoma (NB) via either monoclonal antibodies or more recently CAR-Ts^{6,7}. In particular, GD2-specific CAR-Ts (GD2.CAR-Ts) have been well tolerated and produced clinical benefits in some patients with NB⁶. We evaluated GD2 expression in human RB specimens. GD2 was expressed on the human RB, but not on the normal human retina (Figure 1A). GD2 was also found highly expressed in the two RB cell lines Y79 and WERI-rb-1 (Figure 1B). We then investigated whether GD2.CAR-Ts can target RB cancer cells by co-culturing CAR-Ts with RB tumor cell lines. Both control CD19-specific CAR (CD19.CAR) and GD2.CAR were expressed in transduced T cells (Extended Data Figure 1A), but only GD2.CAR-Ts specifically eliminated both Y79 and WERI-rb-1 RB cells (Figure 1C). The antitumor activity of GD2.CAR-Ts was corroborated by the detection of Th1 cytokines (IFN- γ and IL-2) in culture supernatants as well as by their proliferation in response to RB tumor cells (Extended Data Figure 1B).

Result

To evaluate the antitumor effects of GD2.CAR-Ts *in vivo*, we used in an *in-situ* xenograft nude mice model as previously described⁸. In this model, RB tumor cells are implanted in the subretinal space, creating a tumor that mimic the human disease, which better recapitulates the origin of human RB as compared to intravitreal inoculation. We stably expressed (via retroviral vector) the enhanced green fluorescence protein (eGFP)-firefly luciferase (FFLuc) in the Y79 RB cell line and engrafted 10⁴ cells via subretinal injection. Seven days post-grafting, GD2.CAR-Ts or CD19.CAR-Ts (10⁶ cells) were inoculated intratumor, via subretinal injection. GD2.CAR-Ts promoted delayed tumor growth measured by either bioluminescence (BLI) or direct tumor measurement (Figures 1D, 1E, and 1G), but did not produce tumor eradication as all mice developed tumor and required euthanasia by day 70 (Figures 1F, 1G). Overall, these data suggest that local injection of GD2.CAR-Ts can transiently control RB growth.

In the attempt to eradicate the tumor, we hypothesized that providing GD2.CAR-Ts with self growth factor support would enhance their therapeutic effect by prolonging their persistence within the tumor. We introduced IL-15, an immune-enhancing cytokine known to promote T cell proliferation, activation, and survival⁹ into the GD2.CAR construct¹⁰ (Extended Data Figure 1A). While the addition of IL-15 did not cause RB regression in mice treated with CD19.CAR-Ts, GD2.CAR-Ts releasing IL-15 (GD2.CAR.15) showed significantly improved antitumor effects with 60% treated mice remaining tumor free up to day 70 (Figure 2A-E) ($p=0.0010$). These data indicate that IL-15 by sustaining the local survival of GD2.CAR-Ts enhances their antitumor effects.

Intratumor delivery of T cells occurs as a single “all-at-once” delivery and it does not provide the extended time of intralesional persistence of CAR-Ts required to sustain antitumor effects. A delivery vehicle promoting CAR-T cell persistence would be highly preferred. *In-situ* formed injectable hydrogels allow localized and sustained cell delivery¹¹⁻¹³. We selected the chitosan-polyethylene glycol (PEG) thermosensitive hydrogel forming gel at physiologic temperature studied as optimal injectable hydrogels to deliver CAR-Ts within the eye (Extended Data Figures 2 and 3). The majority of T cells encapsulated within the hydrogel remained viable, were released from the gel within one week *in vitro* and retained their cytotoxic activity (Extended Data Figure 4). We next tested if the encapsulated GD2.CAR.15-Ts would eradicate RB. The chitosan-PEG hydrogel significantly improved the antitumor effects of GD2.CAR-Ts (Figure 3A, $P = 0.0024$, $n = 15$), but the combination of GD2.CAR.15-Ts and chitosan-PEG hydrogel completely controlled the tumor growth and prevented tumor recurrence ($p < 0.0001$, $n = 15$) (Figure 3B-D). Histological study confirmed that the combination of injectable hydrogel and GD2.CAR-Ts or GD2.CAR.15-Ts not only allowed tumor clearance, but also improved structural recovery of the retina (Figure 3E). Analysis of the dissected posterior segments at day 60 post-treatment showed the presence of GD2.CAR-Ts in mice engrafted combining hydrogel and GD2.CAR-Ts secreting IL-15 (Figures 4A and 4B). The retinal structure was essentially preserved, even if signs of inflammatory response and a cavity corresponding to the eliminated tumor were observed (Figure 4C). In addition, the hydrogel facilitated T cell migration and further improved their viability. Western blotting of the lysates obtained from the eye of treated mice further revealed that the hydrogel increased IL-15 detection within the eye. This effect can be due to both enhanced persistence of the T cells and increased retention of IL-15 within the eye (Extended Data Figures 5A and 5B). Overall these data support the conclusion that the local co-injection of the chitosan-PEG thermosensitive hydrogel and GD2.CAR.15-Ts eradicate RB.

To further evaluate if the proposed local immunotherapy preserves the retina function, we performed Fundoscopy, optical coherent tomography (OCT), or Electroretinography (ERG) analysis. The fundus imaging revealed morphological changes under the retina after tumor engrafting and treatment⁸. After Y79-eGFP-FFLuc tumor cell engraftment, the GFP signal was clearly detectable in control mice, but absent in mice treated with the combination of hydrogel and GD2.CAR-Ts secreting IL-15, without evidences of macroscopic damages of the retina (Figure 5A). ERG performed to test the retinal response to light showed loss of retinal response to the light in tumor bearing mice as decrease in both scotopic a and b waves and photopic b wave (Figure 5B). In contrast, in mice treated with the combination of hydrogel and GD2.CAR.15-Ts the retinal function was significantly improved ($p = 0.009$, $n = 7$) (Figure 5B).

Discussion

Here we propose the local immunotherapy approach that combines effector T cells targeting GD2, the T cell growth factor IL-15 and chitosan-PEG injectable hydrogel to eradicate RB in a preclinical model. GD2 is a marker on the neuroepithelia tumors and GD2-specific antibodies and CARs have been developed^{6, 7, 14}. However, no attempt has been so far reported to target GD2 in RB. GD2.CAR-Ts showed potent cytotoxic activity against RB

tumor cell lines *in vitro*. However, GD2.CAR-Ts eradicated the RB *in vivo* only upon the incorporation of IL-15 and encapsulation within the chitosan-PEG injectable hydrogel, which improve T cell lifespan and biodistribution, respectively. While we previously reported the critical role of IL-15 expressed by CAR-Ts in prolonging their survival and antitumor activity in metastatic tumor models^{10, 15}, we report here the critical role of a biomaterial scaffold in supporting the antitumor effect of CAR-Ts when injected intratumor.

The chitosan-based injectable hydrogels are biocompatible and biodegradable^{13, 16, 17}. Previous study using chitosan-PEG hydrogel demonstrated its positive effects on cell delivery due to its low viscosity^{13, 18}. The in-situ gelation we propose facilitates CAR-T cell movements and confines IL-15 within the hydrogel, thus facilitating the use of the cytokine by the CAR-Ts and limiting its diffusion. Remarkably, the immunotherapy approach we propose preserves retinal function. While activated T cells may cause toxicity in the retina, the injectable hydrogel reduced inflammation and retinal detachment by localized delivery of T cells. The proposed approach has potential clinical translation since GD2.CAR-Ts and GD2.CAR-natural killer T cells releasing IL-15 are currently in clinical trials in patients with neuroblastoma (NCT03721068 and NCT03294954), although the safety of intra-tumoral injection will be further evaluated in a future study of retinoblastoma immune therapy. Furthermore, the injectable chitosan-PEG hydrogel can be generated following good laboratory practice or good manufacture practice^{11, 16}.

METHODS

Synthesis and characterization of injectable of hydrogels

The chitosan-PEG hydrogel was synthesized adopting from previous study^{17,19}. 0.2 g PEG and 0.2 g N-hydroxysuccinimide (NHS) were dissolved in 5 mL of DMSO and reacted for 72 hours at room temperature. The solvent was then removed using evaporation. Thereafter, chitosan was dissolved in 2% w/v acetic acid aqueous solution to prepare solution at 2% w/v concentration. Reaction product (NHS-PEG-NHS) was mixed with 2% w/v chitosan solution. The reaction was carried out for 24 hours and product was purified via centrifugal ultrafiltration (10 kDa cut-off molecular weight) against DI H₂O. Finally, the product was mixed with β -glycerophosphate solution to finalize hydrogel precursor for injection. The alginate hydrogel synthesis followed our previous study²⁰. Briefly, 5 mL of 2% alginate aqueous solution was mixed with 5 mL of 2% gelatin aqueous solution in 20 mL vial. Next, 40 mg (1-ethyl-3-(3-dimethylaminopropyl) carbodiimide and 30 mg NHS were added and reacted at ambient temperature for 24 hours. The hydrogel precursor was mixed with the same volume of 0.05 M CaCl₂ solution spontaneously to finish crosslinking. The structure of hydrogel was characterized with ¹H NMR spectroscopy (600 MHz INova spectrometer, D₂O solvent), and the morphology was observed by scanning electron microscopy (Zeiss Supra 25). The rheological test was performed on using TA instrument discovery HR-1 rheometer with 20 mm diameter parallel plate at ambient and physiological temperature. Linear viscoelastic region was determined by performing a strain sweep from 0.1 to 10% strain at 1Hz oscillation frequency. The storage and loss modulus were calculated using linear range (0 - 5% strain) average.

Cell culture, production CAR-Ts and functional assays

All cell lines (Y79 and WERI-rb-1) used in this study were purchased from American Type Culture Collection. Y79 and WERI-rb-1 cell lines were cultured in RPMI 1640 media supplied with 10% FBS, and 1% Glutamax. Generation of GD2.CAR-Ts and GD2.CAR.15-Ts followed the same procedure as previously described¹⁰. Briefly, peripheral blood nononuclear cells (PBMCs) were isolated from buffy coats (Gulf Coast Regional Blood Center) using Lymphoprep (Accurate Chemical and Scientific Corporation). PBMCs were then activated using 1 mg/mL immobilized CD3 (Miltenyi Biotec) and CD28 (BD Biosciences) antibodies. After 3 days, cells were transduced with retroviral supernatants and expanded in complete medium supplemented with IL-7 (10 ng/mL, PeproTech) and IL-15 (5 ng/mL, PeproTech) and expanded by feeding them every 3 to 4 days with fresh complete media and cytokines^{21, 22}, and expression of CAR was analyzed following previous procedure¹⁰. Tumor cell immunophenotyping followed the previous procedure¹⁰. Briefly, the Y79 and WERI-RB-1 cells were harvested and washed with FACS Buffer. Next, the tumor cells were incubated with anti-GD2 antibody (BD bioscience, clone 14.g2a) and isotype for 30 minutes 4°C in the dark. Last, the cells were washed with FACS buffer and suspended in FACS buffer for FACS analysis on a BD LSRFortessa cell analyzer with FACSDiva software. FACS data were processed on FCSEXpress 6 software. The gating strategy is shown in extended data Figure 6A. Isotype was used to gate negative samples. The staining methods was described previously, anti-CD3-APC, anti-CD4-FITC, anti-CD8-PE, anti-CD45-BV510, anti-GD2-BV711 antibodies were used. The counting beads (Count Bright, Invitrogen) were added following the manufacturer's instructions.

T cell release from the hydrogel and functional activity

The alginate and chitosan hydrogels were encapsulated with 0.2 mL volume with 1×10^6 , 4×10^6 , and 1×10^7 T cells. The hydrogel was formed in 6-well transwell plate with culture media. Cell counting and cell viability using trypan blue were performed daily. To measure the antitumor activity of CAR-Ts released from the hydrogel 2.5×10^5 Y79 or WERI-RB-1 cells were seeded on 24-well plate prior to add hydrogel. T cells were encapsulated with alginate-gelatin and chitosan-PEG hydrogels in 0.2 mL volume with 0.5×10^5 or 2.5×10^5 T cells prior to transfer to each well. Blank hydrogel and T cells without hydrogel were used as control groups. After coculture for 72 hours, tumor cells were imaged by fluorescence microscope and enumerated by flow cytometry. Cytokines (IL-2 and IFN- γ) released by CAR-Ts upon targeting tumor cells were measured in the coculture supernatant collected after 24 hours using specific enzyme-linked immunosorbent assays (R&D systems, Inc.).

In vivo studies

Animal experiments were conducted in accordance with the policies of the Institutional Animal Care and Use Committee at the University of North Carolina at Chapel Hill. Male and female 6- to 8- week old NU/NU nude mice were purchased from the Charles River Laboratory. Ketamine (100 mg/kg)/Xylazine (10 mg/kg) injected intraperitoneally or inhalation of 2.5% isoflurane were used for anesthesia. To establish the xenograft RB model, 1×10^4 tumor cells suspended in a final volume of 1 μ L were injected under the retina⁸. Seven days later, 1×10^6 CAR-Ts in phosphate buffered saline (PBS) or encapsulated in 1

uL hydrogel were also injected under the retina. Blank hydrogels were also used as controls. Tumor growth was monitored by BLI using Caliper IVIS imaging system. Mice were infused with 150 mg/kg body D-Luciferin 10 min prior to imaging. The exposure time was 1 or 5 minutes, and BLI was determined using Living Image software. Mice were euthanized when tumor invaded the cornea or tumor volume exceed 500 mm³. The photographs of mice were shown in extended data Figure 6B.

Fundus and OCT

Mice were anesthetized and pupils dilated with 1% tropicamide. Fundus images were captured using a Micron IV retinal imaging system (Phoenix Research Labs). Corneas were moistened with Genteal lubricant eye gel (Novartis) and positioned with the Micron eyepiece in direct contact with the eye through the gel. OCT images were captured using the full-scan setting as an average of 10 frames/scan.

Electroretinography (ERG)

ERG was recorded and analyzed according to previously published protocols²³. Briefly, mice were dark-adapted overnight, and eyes were dilated with 1% tropicamide under dim red light. Genteal was applied to each eye, and recordings were made using gold electrodes that were placed on the cornea. The reference electrode was inserted subcutaneously behind the head, and the ground electrode in the tail. ERG was collected using an Espion E2 system (Diagnosys), and mice body temperature was maintained at 37°C during the experiment.

T cell migration and IL-15 retention assay

The migration of GD2.CAR.15 CAR-Ts was measured by placing 10⁶ T cells in chitosan-PEG hydrogel or PBS into a three-dimensional collagen gel (2 mg/mL, Sigma) in complete medium²⁴. T cells migrated through the collagen gel boundary were stained with LIVE/DEAD viability kits (Invitrogen) and imaged with fluorescence microscopy. The migration distance was measured using ImageJ software.

Histology sectioning, immunohistochemistry, and western blotting.

Human RB tumors were frozen in optimal cutting temperature compound and serially sectioned (10 µm). Slides were fixed with 4% PFA for 10 minutes (−20°C) and immunohistochemical analyses were performed. RB tumors were detected with anti-GD2 (1:400, MilliporeSigma) antibody. Alexa Fluor 488 anti-mouse (Life Technologies) were used as secondary antibodies respectively (1:500). Omission of the primary antibody was used as a negative control for nonspecific binding. Cell nuclei were stained for 10 minutes with 4',6'-diamidino-2-phenylindole (DAPI, 1: 5,000) for confocal microscopy. Mice were randomly selected from each cohort and sacrificed for histology and western blotting analysis 35 days post treatment. The haemotoxylin and eosin (H&E) staining of eye tissues were carried out by the UNC Center for Gastrointestinal Biology and Disease Histology Core Facility. Briefly, fixed tissues were embedded in paraffin and sectioned into 10-mm sections for eye tissues. Sections were stained with H&E according to conventional staining protocols and visualized using conventional bright-field microscopy. Mouse eyes were dissected. Lysate obtained from the dissected eye were used for western blotting. Lysates

were loaded in sample buffer and separated using 4-20% SDS-PAGE gels (Bio-Rad). Blots were transferred, stained (anti-IL-15 antibody, ProSci 1:500), and imaged according to the manufacturer's (Bio-Rad) instructions.

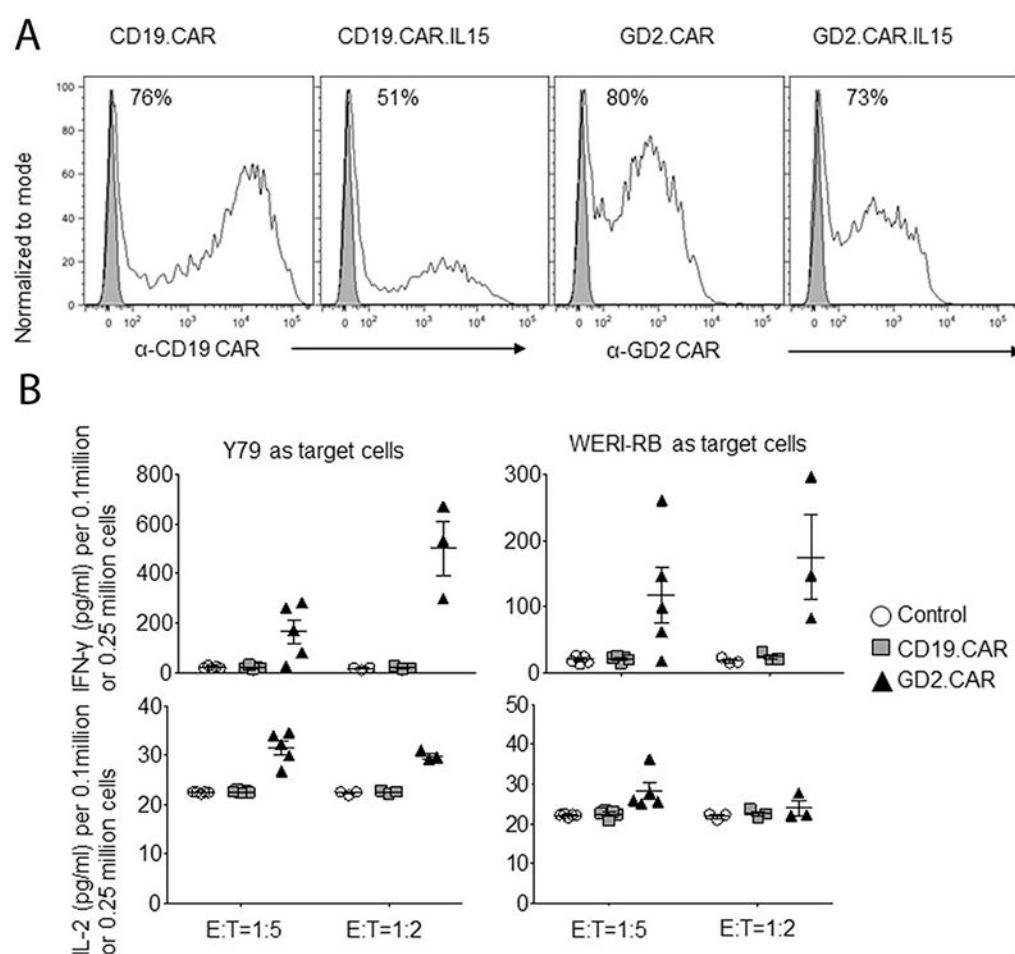
Statistics and reproducibility

All quantitative data were expressed as the mean and standard deviation. The size of animal samples was calculated based on GPower7.1 software and logistic limitations, animals were randomly allocated into different groups. All the studies are sufficiently powered to yield statistically significant results (95% confidence interval). The data were analyzed by one-way or two-way analysis of variance (ANOVA), and a p-value less than 0.05 was considered significant. Survival curves were plotted using the Kaplan-Meier method, and the differences in the survival between groups were assessed by log-rank test.

Reporting Summary

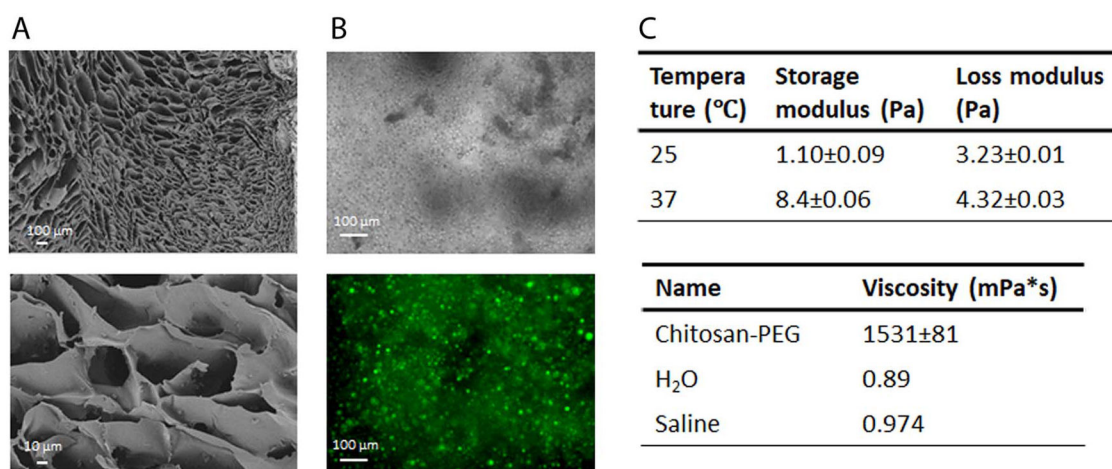
Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Extended Data



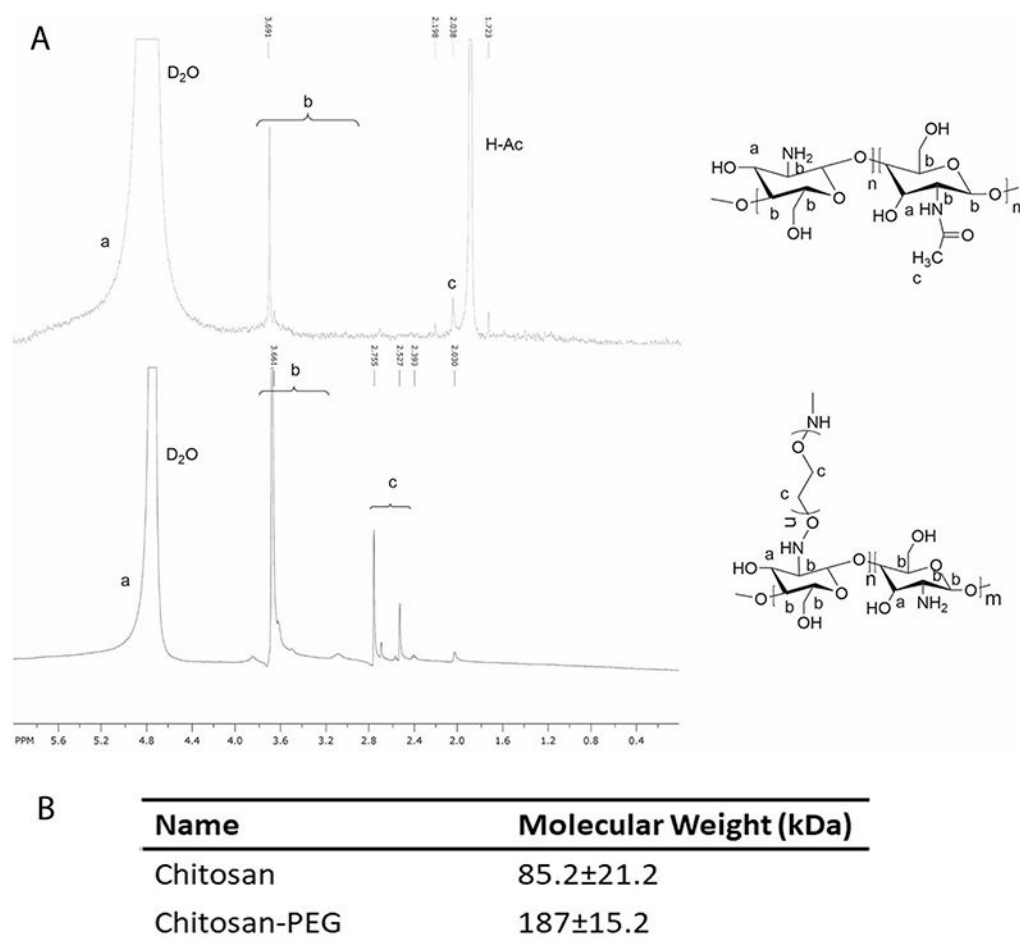
Extended Data Fig. 1. T cells expressed CAR and effector-tumor coculture exhibited elevated cytokine release

(A) CAR expression in T cells transduced with retroviral vectors encoding GD2.CAR, CD19.CAR, GD2.CAR.IL15, and CD19.CAR.IL15. CAR expression was evaluated by flow cytometry. (B) IL-2 and IFN- γ released in the culture supernatants of CAR-Ts co-cultured with tumor cells. Supernatant was collected at 24 hours and cytokines were measured by specific ELISA (E:T=1:5, n = 5 donor/group, E:T=1:2, n = 3 donor/group). Data presented show mean and SEM.



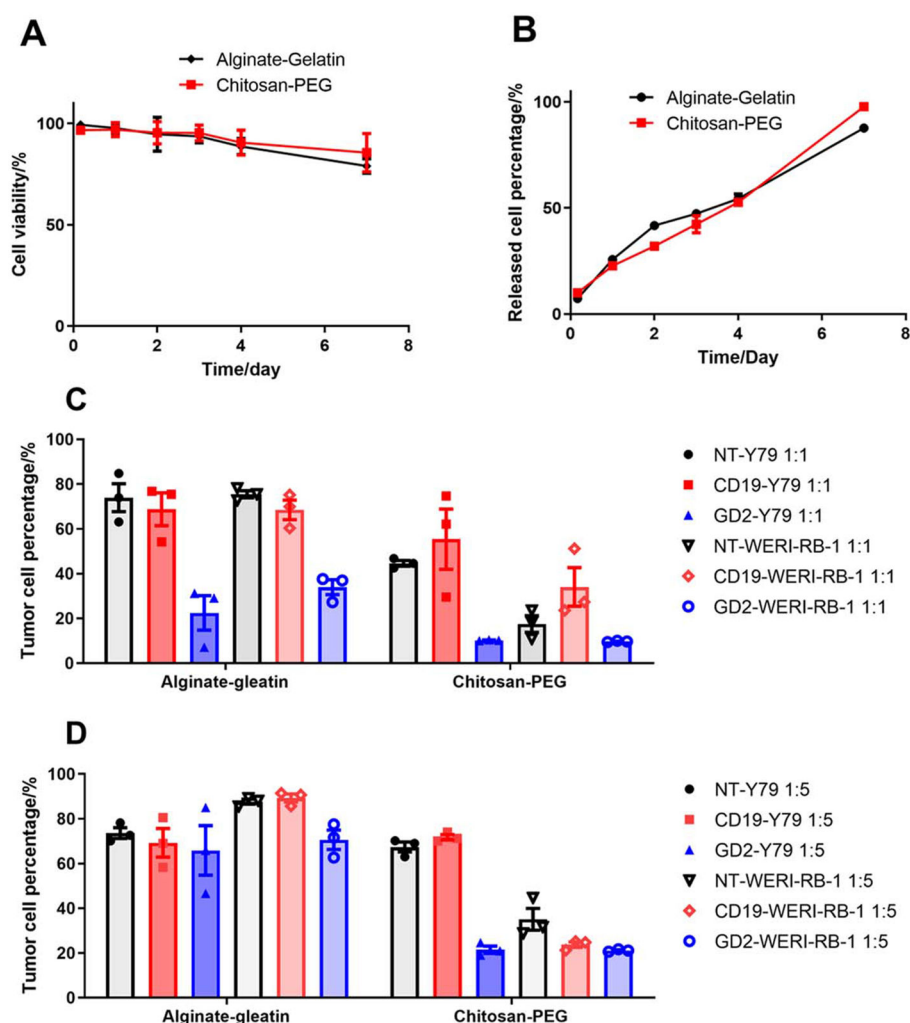
Extended Data Fig. 2. The morphologic and rheologic properties of Chitosan-PEG hydrogel

(A) Lyophilized injectable hydrogel was imaged by scanning electron microscopy. The image showed 63 μ m average pore diameter of the hydrogel (n=10 technical replicates). (B) GD2.CAR-Ts labelled with eGFP-FFLuc were encapsulated into the hydrogel and visualized by fluorescence microscopy (B). Assessment of the viscoelastic properties of the injectable hydrogel tested at 23°C and 37°C (n=10 technical replicates) (C). Rheological test result of the chitosan-PEG injectable hydrogel and water/saline¹⁶ (n=10 technical replicates). Data presented show mean and SD.



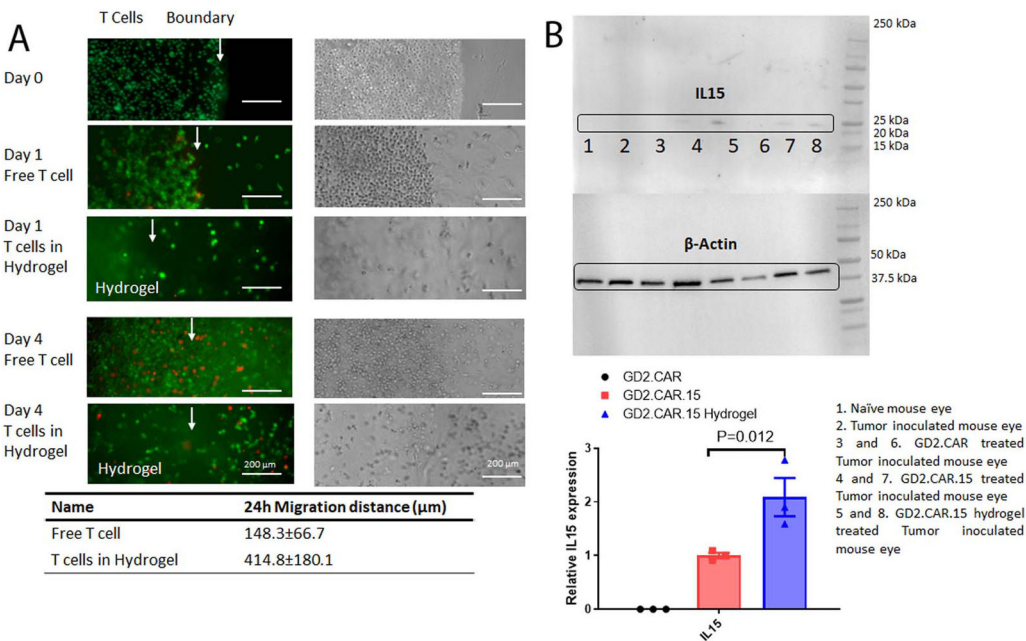
Extended Data Fig. 3. Molecular characterization of chitosan-PEG hydrogel

(A) ^1H NMR spectra of the chitosan and the chitosan-PEG hydrogel precursor and (B) molecular weight of Chitosan-PEG precursors measured by dynamics light scattering ($n=12$ technical replicates). Data presented show mean and SD.

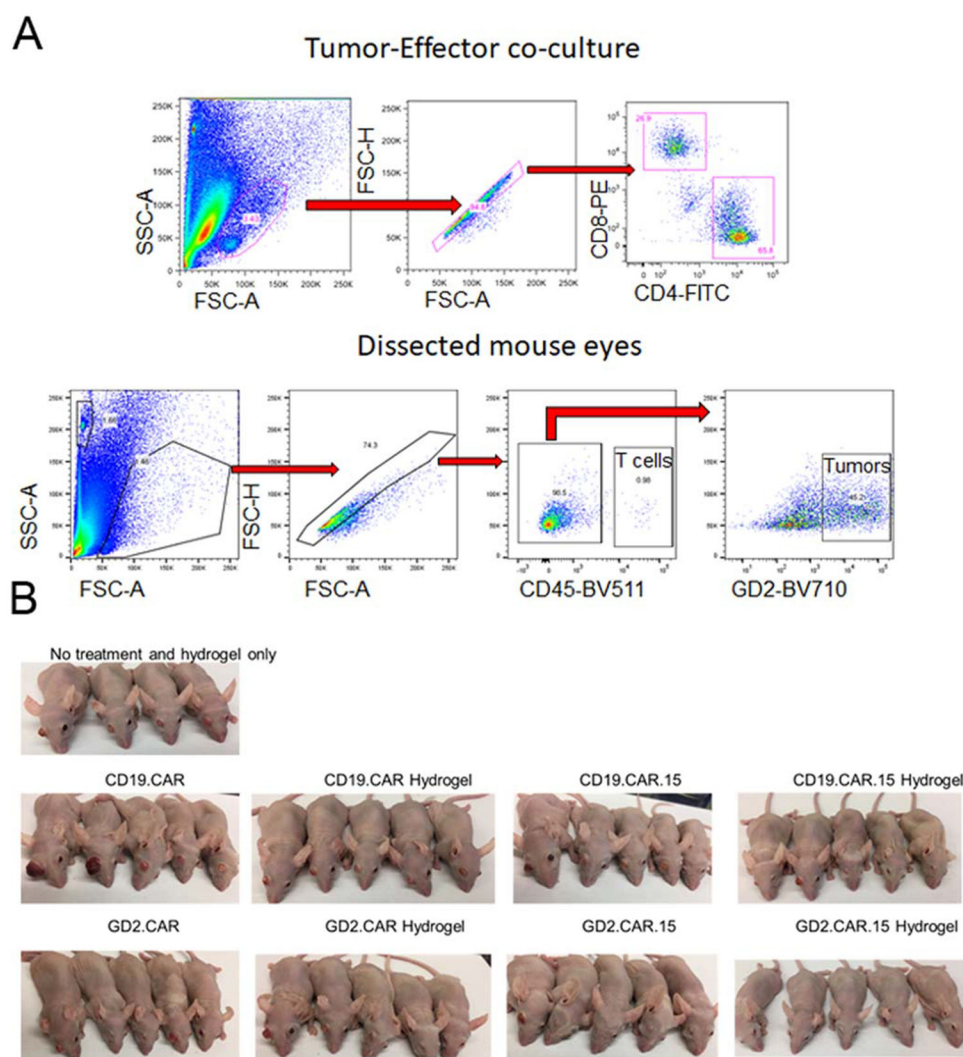


Extended Data Fig. 4. CAR-Ts release from hydrogels and chitosan-PEG hydrogel encapsulated GD2.CAR-Ts effectively eliminated tumor

(A) Viability of CAR-Ts released from alginate-gelatin and chitosan-PEG injectable hydrogel (n =3 donor/group). (B) CAR-T cell release Profile from alginate-gelatin and chitosan-PEG injectable hydrogel (n =3 donors/group). RB tumor cell lines were co-cultured with GD2.CAR-Ts or CD19.CAR-Ts at the effector to tumor ratio of 1:1 (C) and 1:5 (D) for 7 days upon encapsulation with the hydrogels. All cells were collected and analyzed by flow cytometry to enumerate T cells and residual tumor cells in the culture (n =3 donors/group). Data presented show mean and SEM.



Extended Data Fig. 5. The hydrogel facilitates the antitumor activity of GD2.CAR-Ts. (A) Migration assay showing T cell migration and viability in the collagen gel (Fluorescence Green: Living cells, Red: Dead cells) on days 1 and 4. ²⁴ (B) Western blotting analysis of the protein lysate of the dissected eyes at day 35 post-treatment to detect IL-15. The statistical significance was determined using one-way ANOVA with Tukey's post hoc test (GD2.CAR.15 versus GD2.CAR.15 hydrogel) P=0.012. n=3 individual sample/group. Data presented as mean and SEM.



Extended Data Fig. 6. The Gating strategy and photography of mice

(A) Gating strategy to assess effector-tumor coculture (CD4⁺/CD8⁺ cells) and the presence of T cells (CD45⁺ cells) and RB tumor cells (GD2⁺ cells) in the posterior segment of the eye by flow cytometry. (B) Photographs of representative nu/nu mice (n=15/group) 35 days after engraftment with the Y79-eGFP-FFLuc tumor cell line and receiving the indicated treatments.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Data availability

Source data for Figure 1-5 and Extended Data Figures 1, 4, and 5 have been provided as Source Data Files. Bioluminescence and imaging data can be found at figshare https://figshare.com/articles/IVIS1_tif/12485699. All other data supporting the findings of this study are available from the corresponding author on reasonable request.

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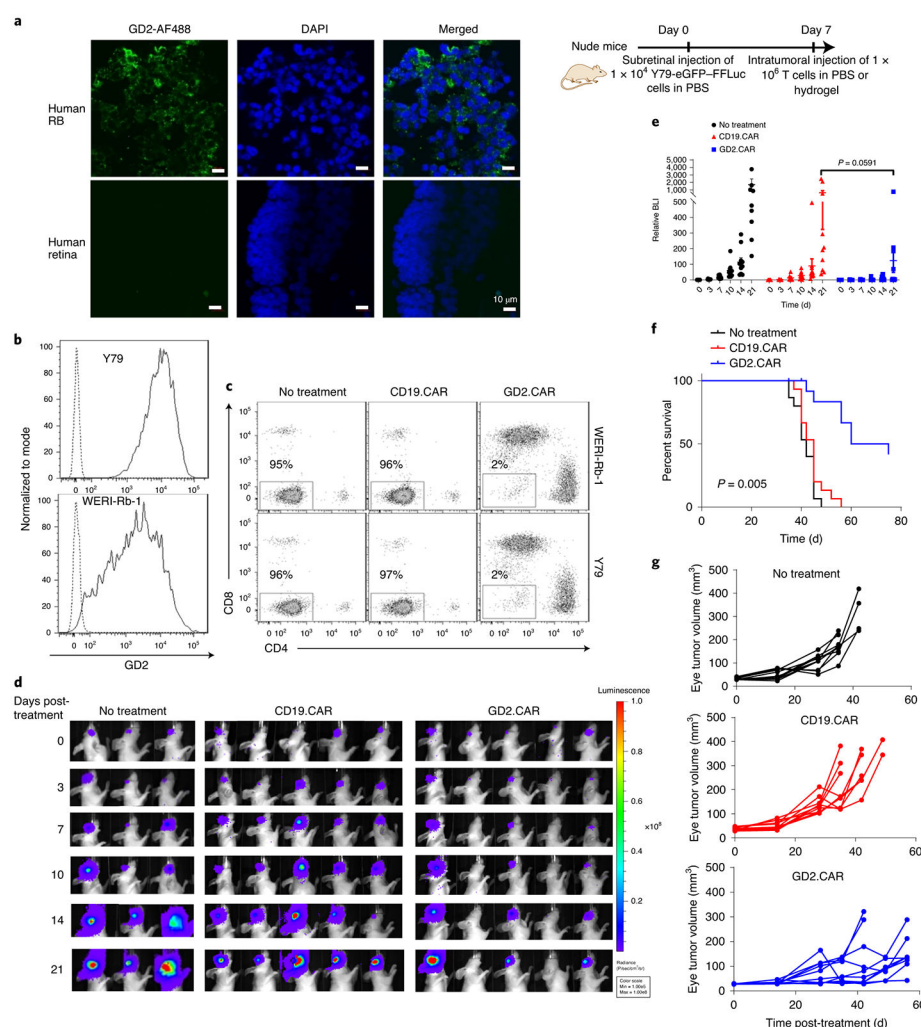


Figure 1. Retinoblastoma express GD2 on the cell surface, and GD2.CAR-Ts control the growth of RB, but fail to prevent tumor recurrence

(A) Representative human RB section stained with the GD2-AlexaFluor488 Antibody showed extensive expression of GD2 on the tumor surface, while normal retina was used as negative control where GD2 is absent (Representative of $n = 3$ individual repeated experiments). (B) Representative immunophenotyping result of Y79 and WERI-rb-1 human RB cell lines revealed expression of GD2 (Y79 and WERI-rb-1, representative of $n = 3$ individual experiments per group). (C) RB tumor cell lines were co-cultured with GD2.CAR-Ts and CD19.CAR-Ts at the tumor to T cell ratio of 1:1. All the cells were collected and analyzed by FACS to measure CD4⁺/CD8⁺ T cells and residual tumor cells (CD4⁻CD8⁻) in the culture (No Treatment, GD2.CAR, and CD19.CAR, representative of $n = 6$ donors per group). (D) Nude mice (Representative of No treatment $n = 8$, CD19.CAR and GD2.CAR $n = 15$ animals per group) were engrafted subretinally with the Y79-eGFP-FFLuc tumor cell line. Seven days later mice were subretinal intra-tumor injected with GD2.CAR-Ts, or control CD19.CAR-Ts. Tumor growth within the eye was imaged and measured by *in vivo* bioluminescence by IVIS. Bioluminescence and imaging data can be found at figshare https://figshare.com/articles/IVIS1_tif/12485699. (E) BLI summary

(Quantification of the same cohort from panel D. Statistical significance was determined by two-way ANOVA with Bonferroni post hoc tests, (CD19.CAR versus GD2.CAR, Day 21) $P = 0.0591$, No Treatment $n = 8$, CD19.CAR, and GD2.CAR, $n = 15$ animals per group). Data presented in mean and SEM. (F) survival curves post treatment (No treatment $n = 8$, CD19.CAR, and GD2.CAR, $n = 15$ animals per group (Quantification of the same cohort from panel D and E). Statistical significance was determined by log-rank test (Two-sided), $P = 0.005$), and (G) tumor sizes measured (No Treatment $n = 8$, CD19.CAR, and GD2.CAR, $n = 10$ animals per group). The relevant source data are presented in figure 1.xlsx.

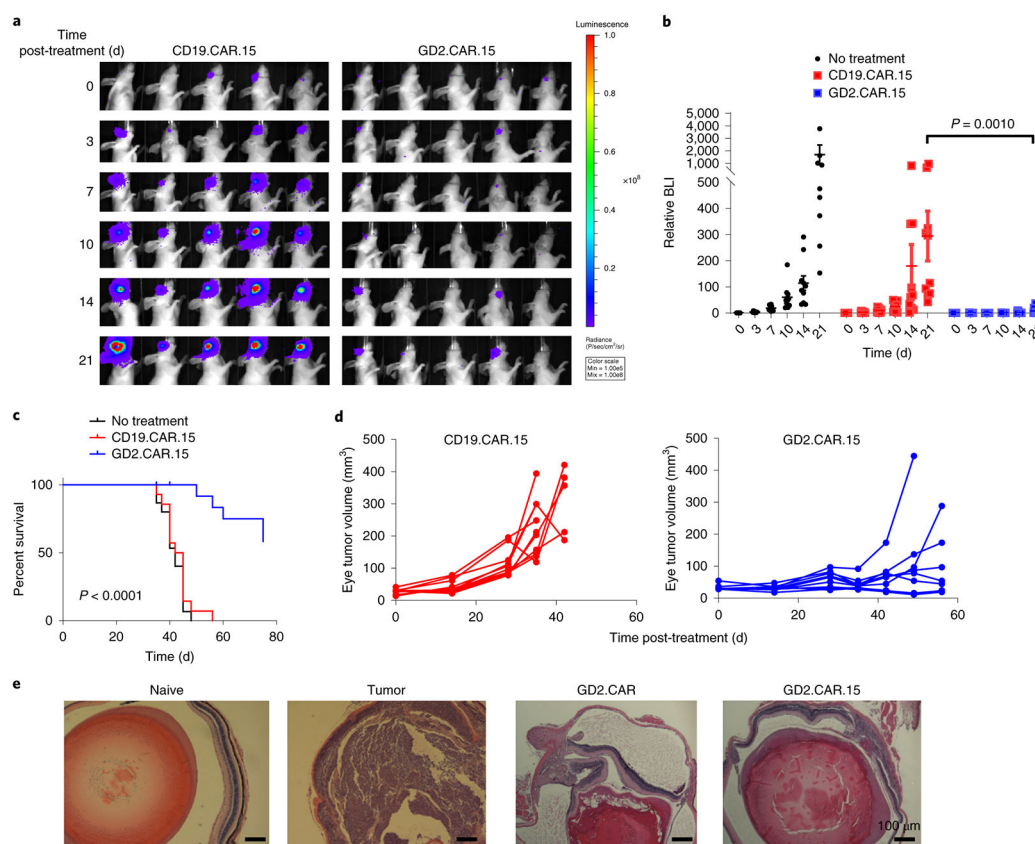


Figure 2. GD2.CAR.15-Ts improve RB treatment but fail to achieve complete tumor eradication.

(A) Nude mice (CD19.CAR.15 and GD2.CAR.15, representative $n = 15$ animals per group) were engrafted intra-retina with Y79-GFP-FFLuc tumor cells. Seven days later mice were intra-tumor injected with GD2.CAR.15-Ts, or control CD19.CAR.15-Ts. Tumor growth within the eye was imaged and measured by *in vivo* bioluminescence by IVIS.

Bioluminescence and imaging data can be found at figshare <https://figshare.com/articles/IVIS1.tif/12485699> (B) BLI summary (Quantification of the same cohort from panel A. Statistical significance was determined by two-way ANOVA with Bonferroni post hoc test (CD19.CAR.15 versus GD2.CAR.15, day 21) $P = 0.0010$, No Treatment $n = 8$,

CD19.CAR.15 and GD2.CAR.15, $n = 15$ animals per group). Data presented in mean and SEM. (C) survival curves post treatment (Quantification of the same cohort from panel A and B. No Treatment $n = 8$, CD19.CAR.15 and GD2.CAR.15, $n = 15$ animals per group).

Statistical significance was determined by log-rank test (Two-sided), $P < 0.0001$, and (D) tumor sizes (CD19.CAR.15 and GD2.CAR.15, $n = 10$ animals per group). (E) H&E paraffin histological section of a representative naïve eye, untreated Y79-GFP-FFLuc engrafted eye, and Y79-GFP-FFLuc engrafted eyes treated with GD2.CAR-Ts or GD2.CAR.15-Ts.

Pictures are all taken 35 days post treatment ($n = 3$ individual experiments). The relevant source data are presented in figure 2.xlsx.

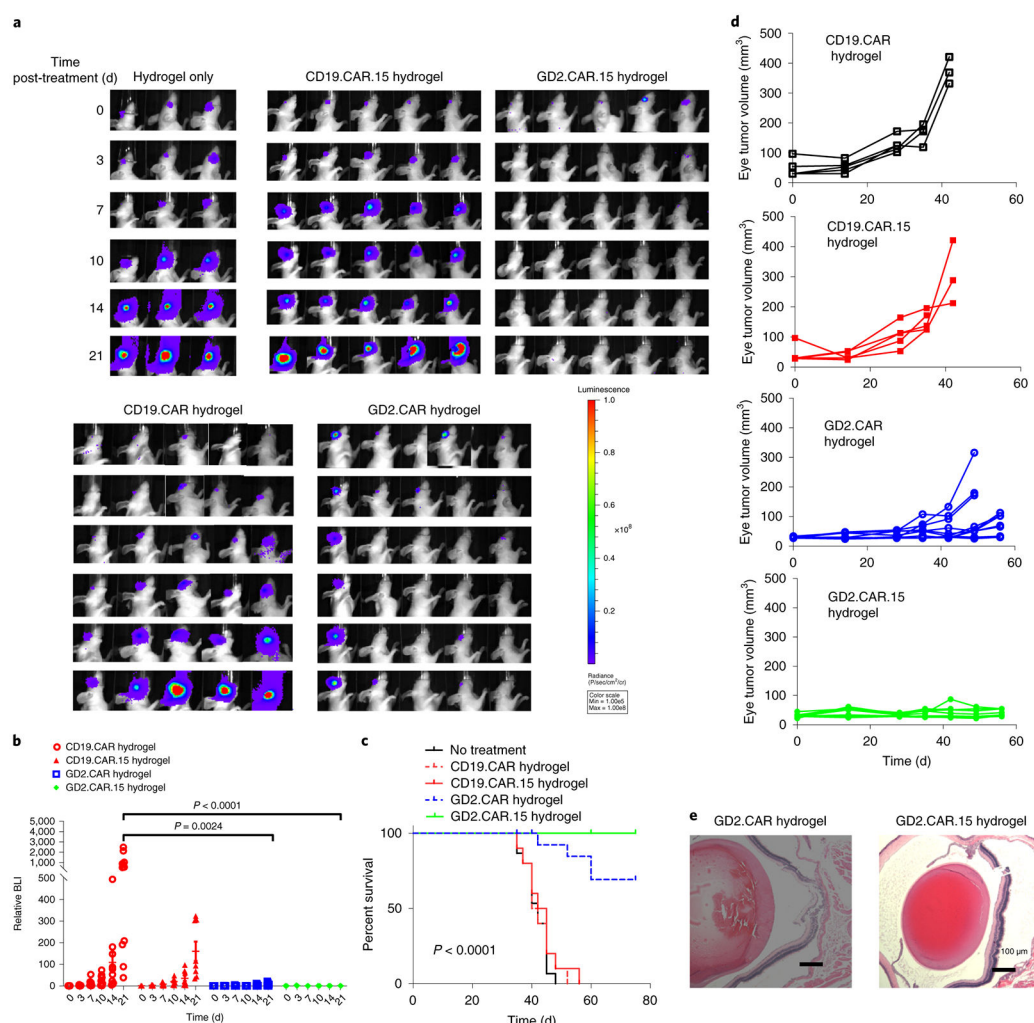


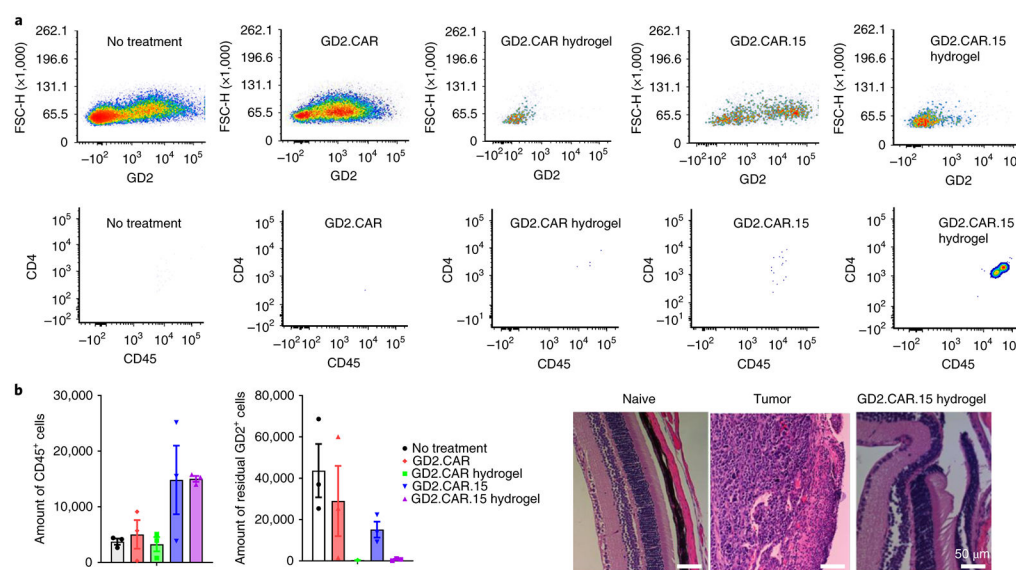
Figure 3. Injectable hydrogel further improved GD2.CAR.15-Ts treatment and prevent tumor recurrence

(A) Nude mice (Representative of hydrogel with CD19.CAR and CD19.CAR.15 $n = 10$ animals per group, Hydrogel with GD2.CAR and GD2.CAR.15 $n = 15$ animals per group, Hydrogel only animal $n = 7$) were engrafted intra-retina with Y79-eGFP-FFLuc tumor cell line. Seven days later mice were intra-tumor injected with GD2.CAR-Ts, GD2.CAR.15-Ts, CD19.CAR-Ts or CD19.CAR.15-Ts encapsulated in chitosan-PEG hydrogel.

Bioluminescence and imaging data can be found at figshare https://figshare.com/articles/IVIS1_tif/12485699 (B) BLI summary (Quantification of the same cohort from panel A.

Statistical significance was determined by two-way ANOVA with Bonferroni post hoc tests (CD19.CAR Hydrogel versus GD2.CAR Hydrogel, Day 21) $P = 0.0024$, (CD19.CAR.15 Hydrogel versus GD2.CAR.15 Hydrogel, Day 21) $P < 0.0001$, Hydrogel with CD19.CAR and CD19.CAR.15 $n = 10$ animals per group, Hydrogel with GD2.CAR and GD2.CAR.15 $n = 15$ animals per group, Hydrogel only $n = 7$ animals per group). Data presented in mean and SEM. (C) survival curves post treatment (Quantification of the same cohort from panel A and B) No treatment $n = 15$, Hydrogel with CD19.CAR and CD19.CAR.15 $n = 10$ animals per group, Hydrogel with GD2.CAR and GD2.CAR.15 $n = 15$ animals per group.

Statistical significance was determined by log-rank test (Two-sided), $P < 0.0001$), and (D) tumor sizes (Hydrogel with CD19.CAR, GD2.CAR, CD19.CAR.15, and GD2.CAR.15, $n = 10$ animals per group). (E) Representative H&E paraffin histological section of Y79-GFP-FFLuc engrafted eyes treated with GD2.CAR-Ts or GD2.CAR.15-Ts encapsulated in hydrogels 35 days post CAR-Ts treatment ($n = 3$ individual experiment). The relevant source data are presented in figure 3.xlsx.



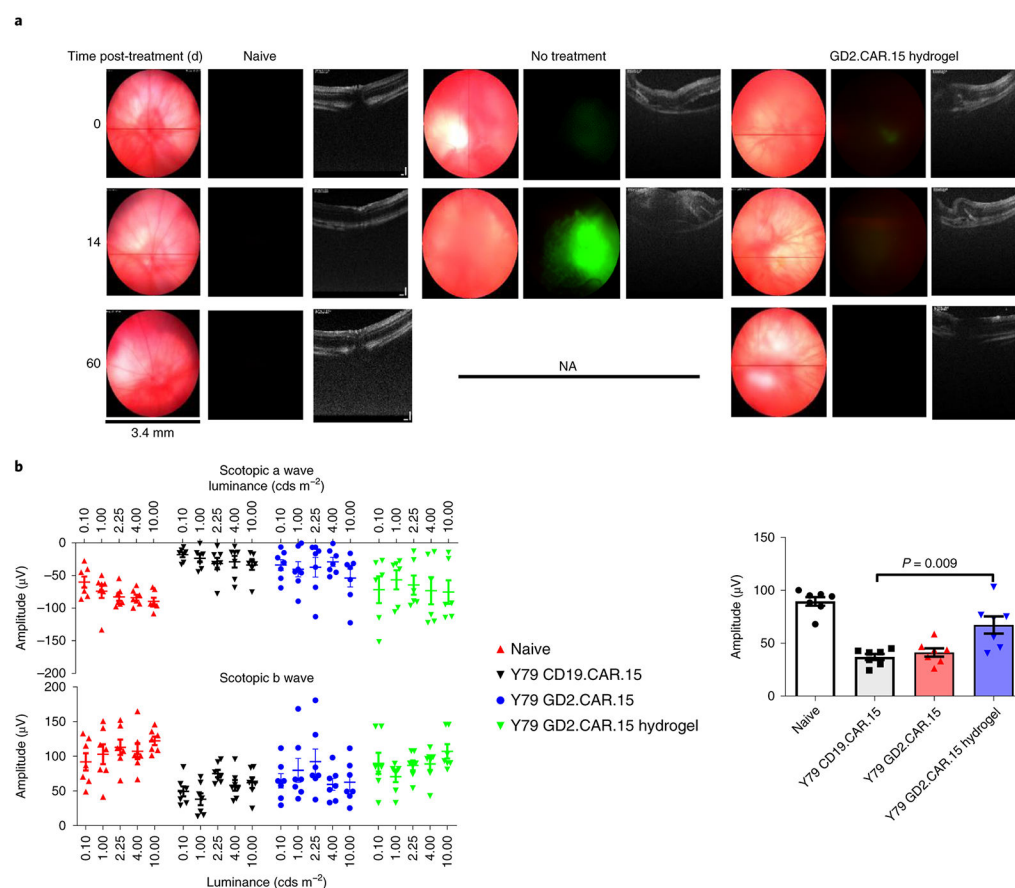


Figure 5. GD2.CAR.15-Ts and hydrogel treatment rescue mouse vision in the absence of toxicity to the retina

(A) Representative Fundus bright field, green fluorescence, and OCT images of naïve, untreated, or GD2.CAR.15-Ts and hydrogel treated nude mice engrafted with tumor cell lines 7 days and 21 days post-grafting ($n = 5$ eyes per group). (B) ERG scotopic a-wave, b-wave, and photopic b-wave of naïve, untreated, GD2CAR.15-Ts and hydrogel treated mice engrafted with tumor cell lines (Naïve, CD19.CAR.15, GD2.CAR.15, and Hydrogel with GD2.CAR.15, $n = 7$ eyes per group). Statistical significance was determined by one-way ANOVA with Tukey's post hoc test (CD19.CAR.15 versus GD2.CAR.15 hydrogel $P = 0.009$). Data presented as mean and SEM. The relevant source data are presented in figure 5.xlsx.